Dataset Optimiser

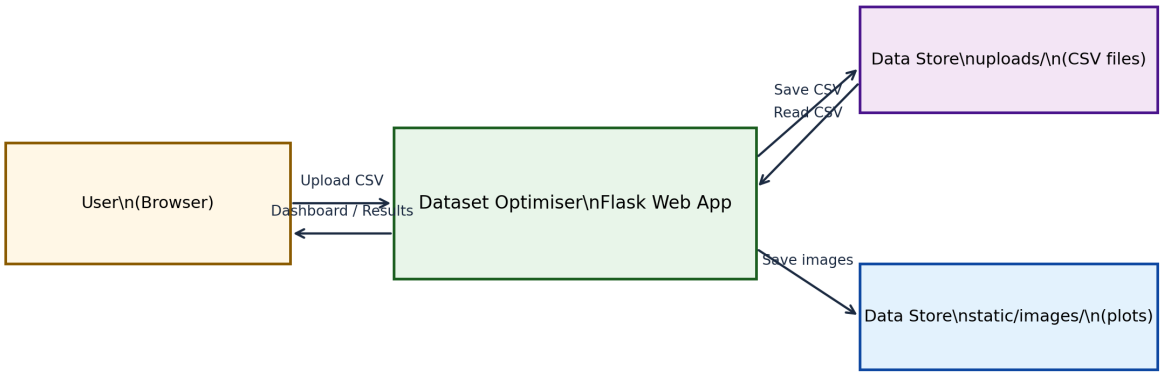
Clear & concise DFD and UML diagrams (examiner-friendly).

Project summary: A user uploads a CSV, the app analyzes it, can clean/optimize/evaluate, then lets the user download results.

DFD Level 0 — Context

Shows the whole system as a single process with external user and file stores.

DFD Level 0 (Context Diagram)

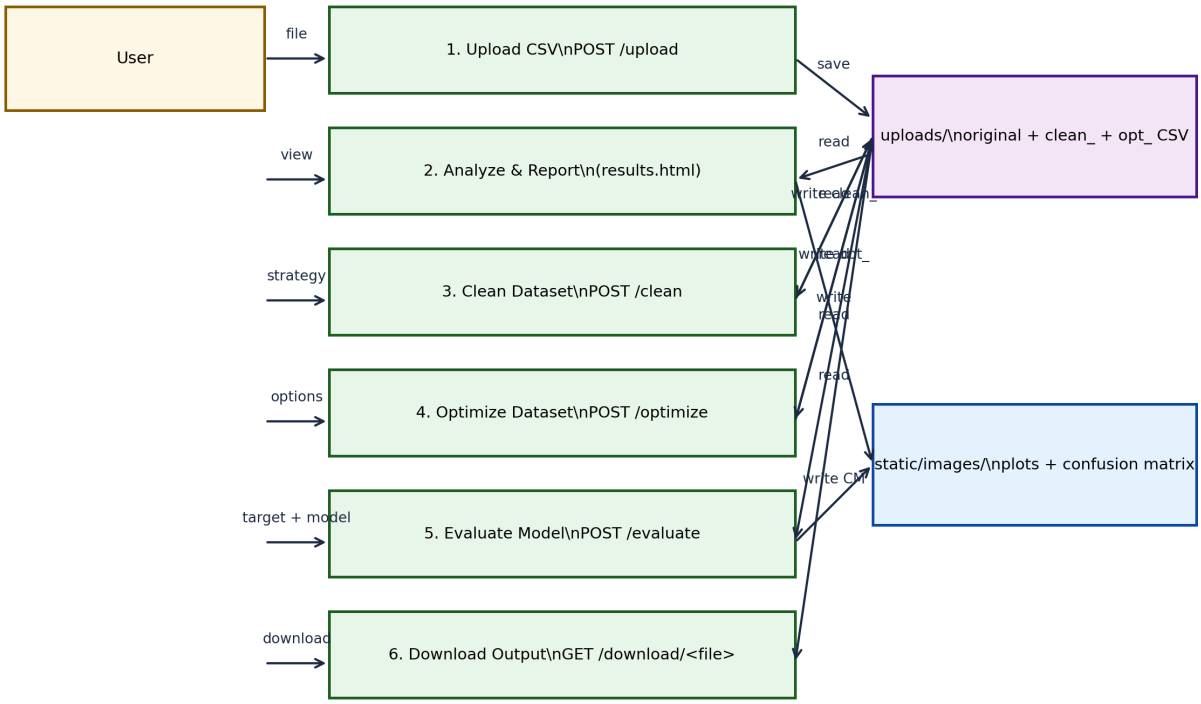


System boundary: everything inside the Flask app.

DFD Level 1 — Main Processes

Breaks the system into the main features: upload, analyze, clean, optimize, evaluate, download.

DFD Level 1 (Main Processes)

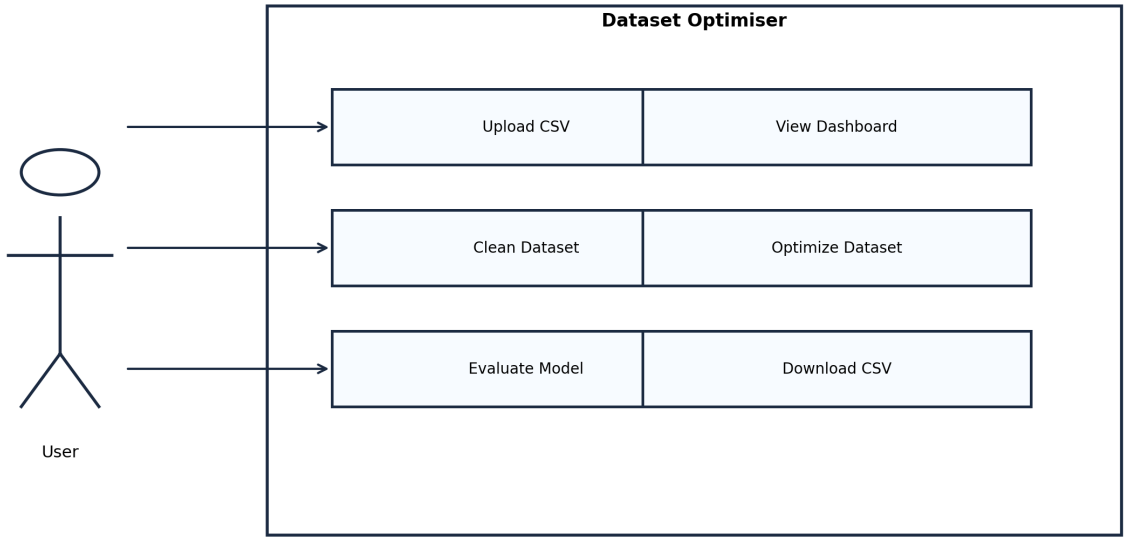


Tip: You can explain this as “Upload → Analyze → (Optional) Clean/Optimize/Evaluate → Download”.

UML Use Case

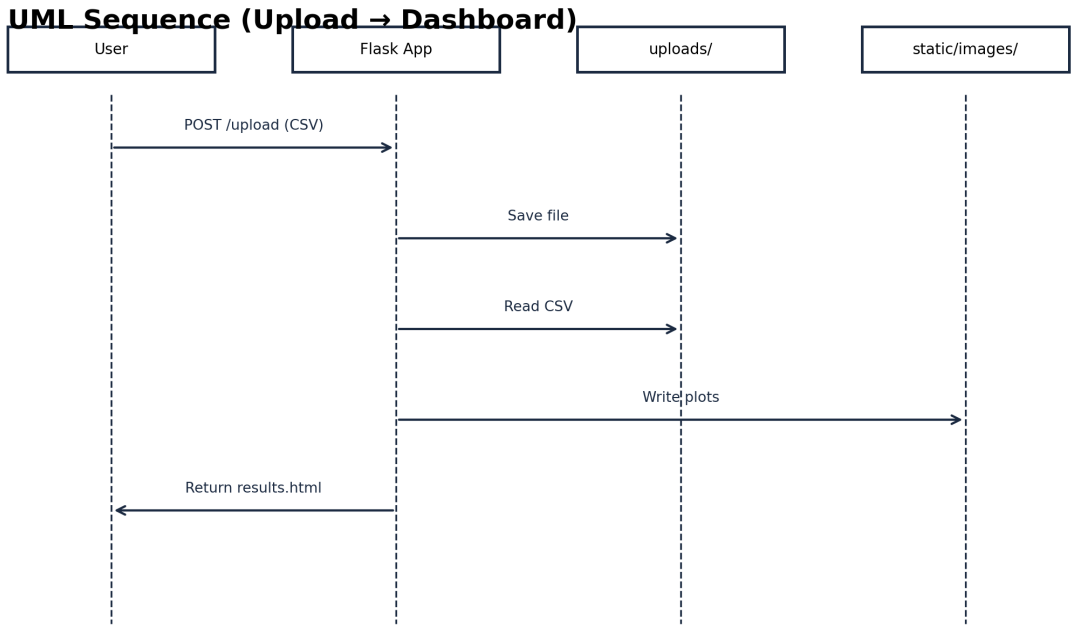
Explains what the user can do in the system (features from the user’s point of view).

UML Use Case (What the user can do)



UML Sequence — Upload to Dashboard

Shows the step-by-step interaction when uploading a dataset and viewing the dashboard.

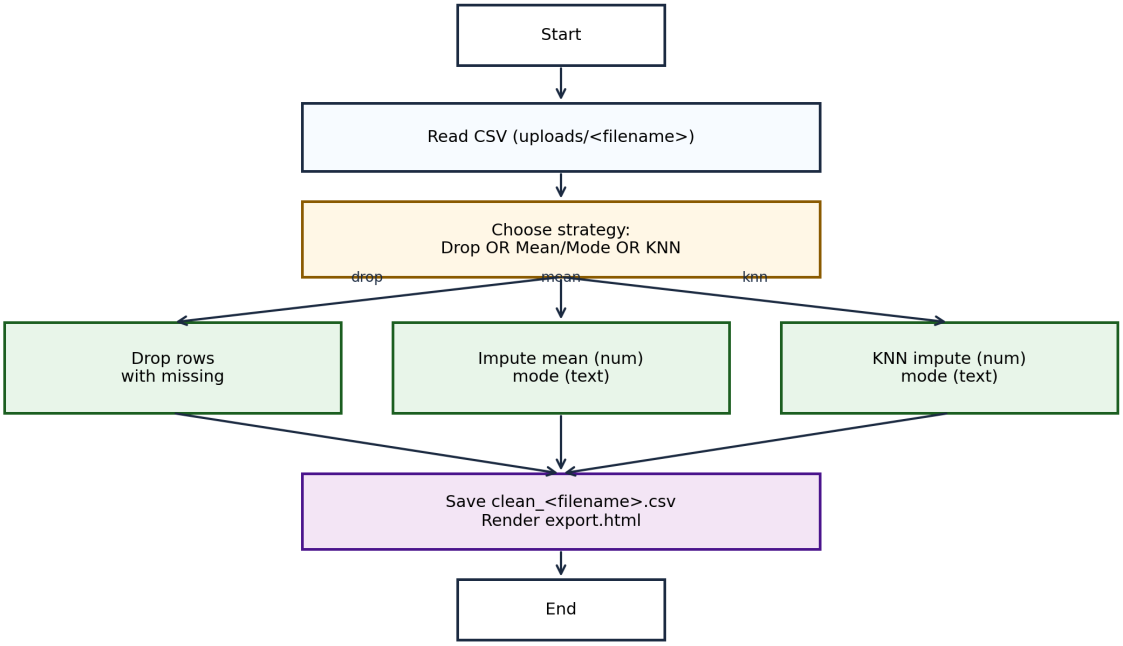


Explain in words: The user uploads CSV → the server saves it → reads it → generates plots → shows the dashboard.

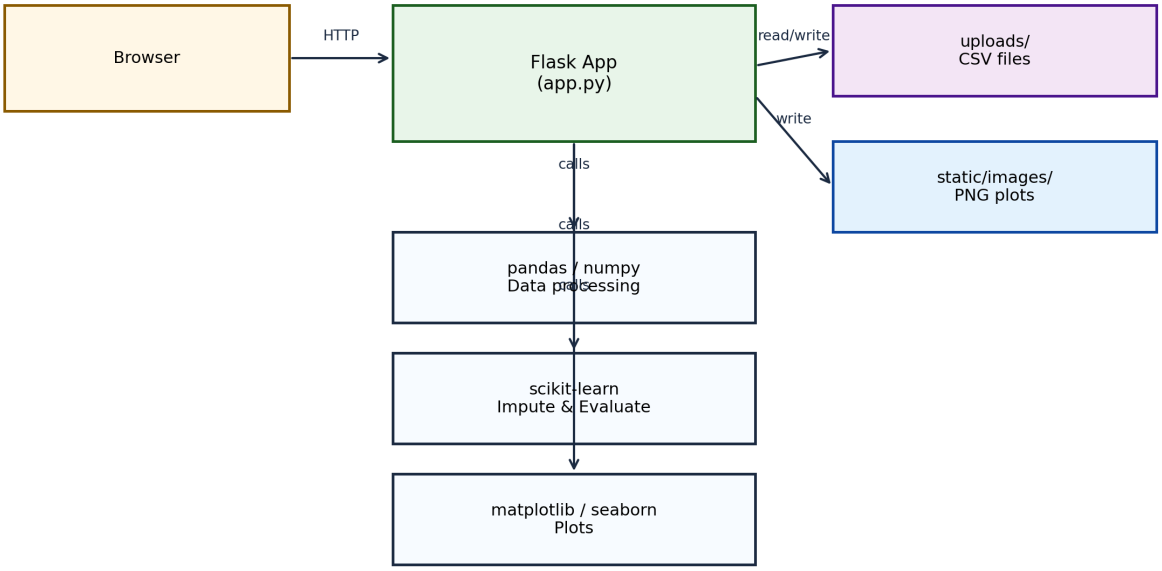
UML Activity — Cleaning

Shows the cleaning logic: choose a strategy → apply it → save cleaned CSV → show download.

UML Activity (Cleaning)



UML Component — Architecture

Shows the main parts used by the app: Flask routes + data processing + ML + plotting + file stores.

UML Component (High-Level Architecture)